Lender Matching Analysis

Optimizing revenue per application through intelligent customer-lender routing

11.0%

Overall Approval Rate

$26.42

Revenue per Application

+33%

Revenue Lift

Financial Services – 100,000 Applications Analyzed – By Lana Marcelin

Three questions. One bottom line.

1

Variable Analysis

Which factors most reliably predict whether a customer will be approved for a personal loan?

2

Lender Profiles

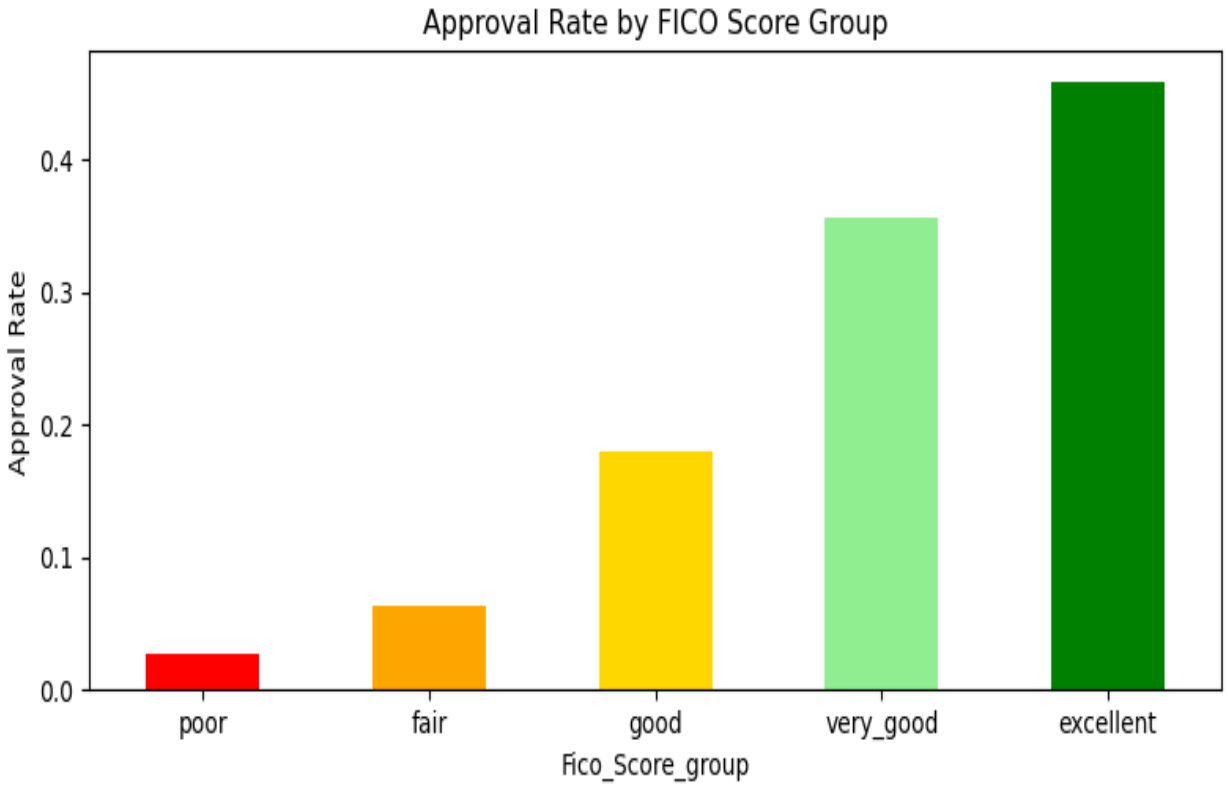
How do Lenders A, B, and C differ in who they approve and what does each lender's ideal customer look like?

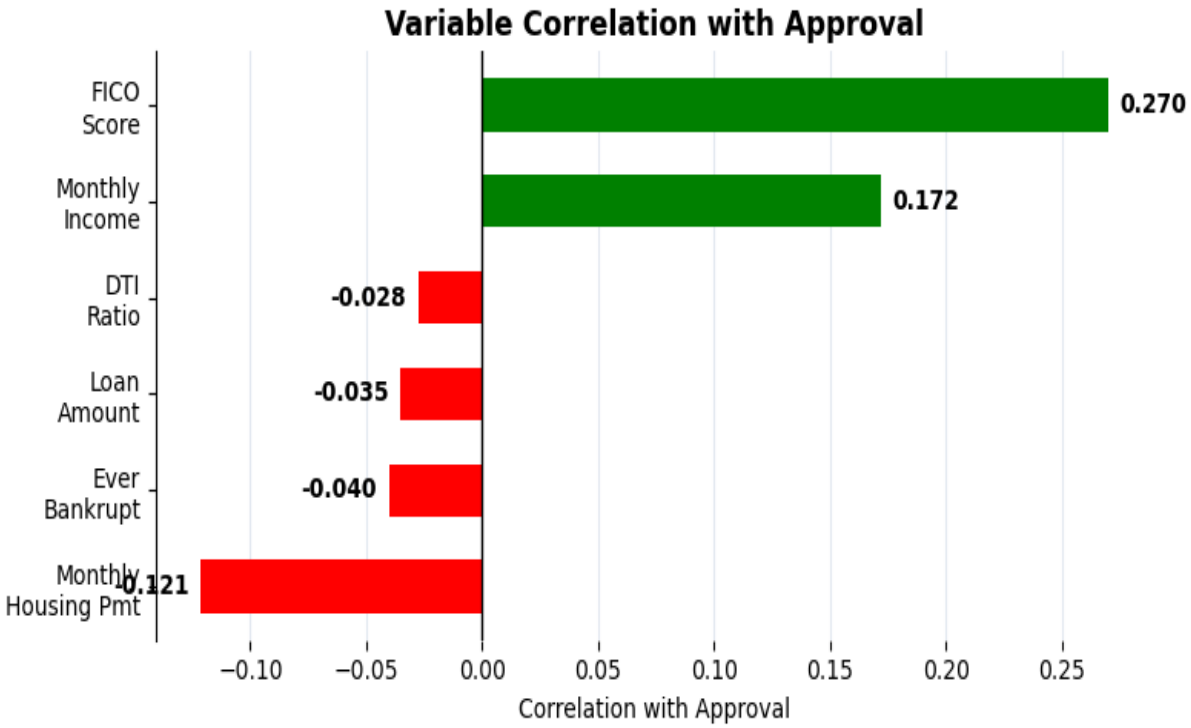
3

Revenue Optimization

How much incremental revenue can Bankrate capture by routing each customer to their best-fit lender?

FICO score and income are the strongest predictors. DTI ratio is the best engineered feature.



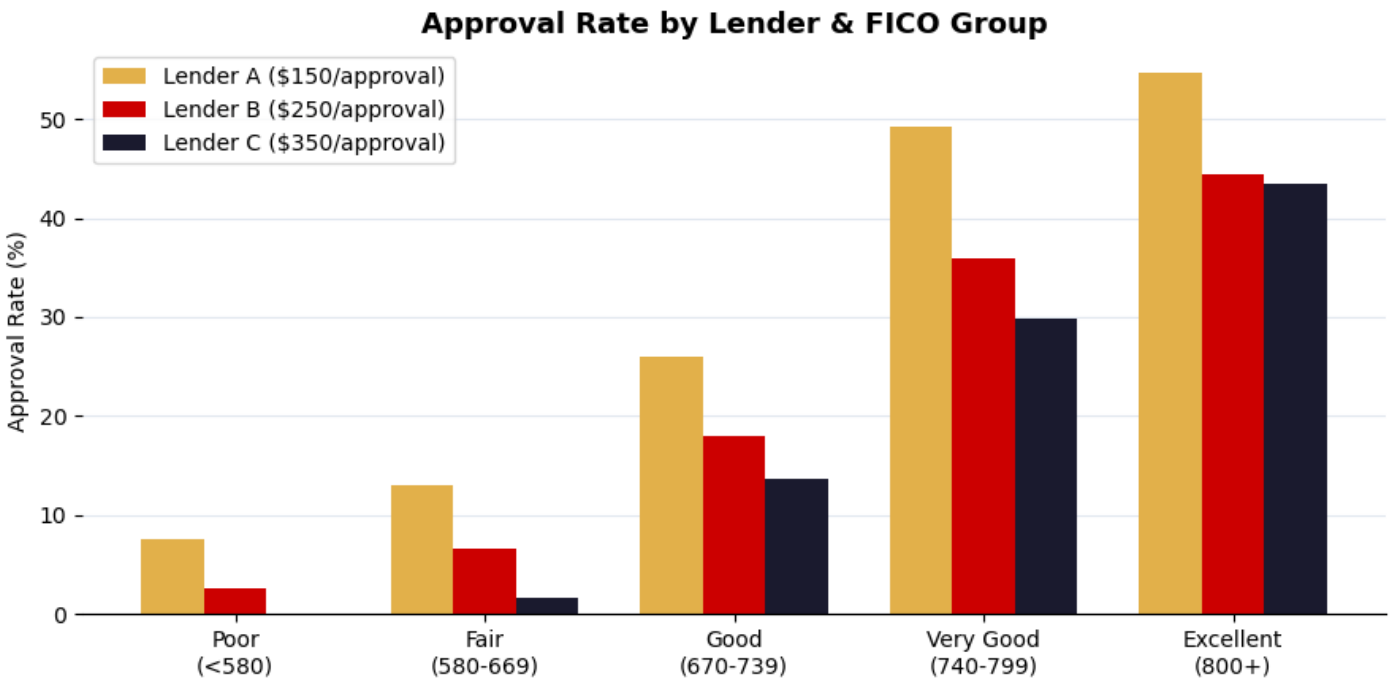


FICO Score
R = 0.27
Strongest raw predictor.

DTI Ratio
R = -0.12
Housing pmt / income

Bankruptcy
3.7% vs 11.1%
Bankruptcy cuts approval rate by 67%

Lender A is most lenient. Lender C pays most but approves only strong profiles.



Lender A

$150/appr ~ 17.1% approval ~ Avg FICO: 675

Most lenient. Best fit for fair/poor FICO customers.

Lender B

$250/appr ~ 11.0% approval ~ Avg FICO: 697

Mid-tier. Balance across segments.

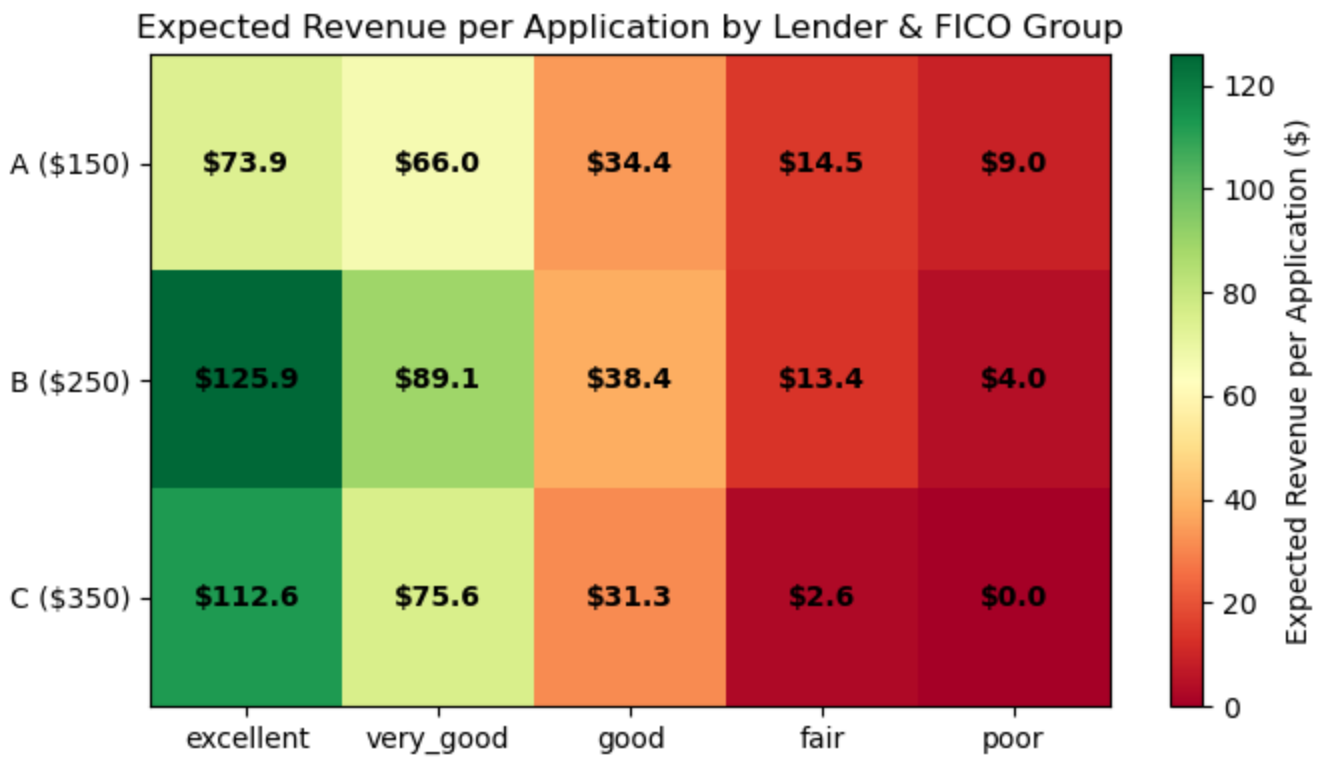
Lender C

$350/appr ~ 7.1% approval ~ Avg FICO: 732

Most selective. Only approves strong profiles. Never approves poor FICO.

- Lender C approval rate for 'poor' FICO = 0.0%. No value in routing low-FICO customers there. Lender A approves poor FICO at 7.6% which is 3x higher than B.

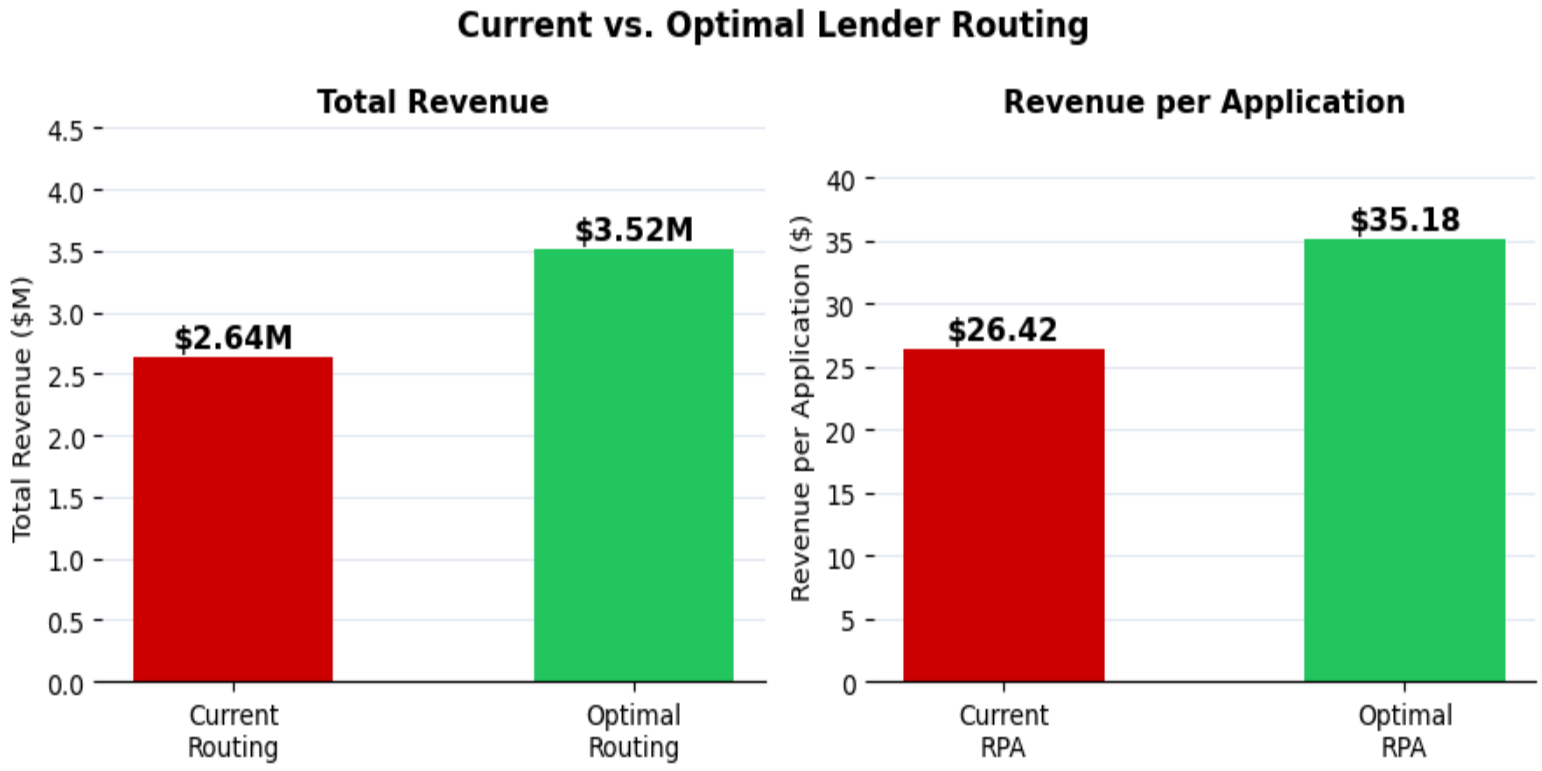
Expected revenue per application = Approval Rate x Bounty. C wins for excellent; A wins for poor/fair.



Optimal Routing Rules

Excellent/Very Good (full-time)	--> Lender C
Excellent/Very Good (unemployed)	--> Lender B
Good (full-time)	--> Lender C
Good (unemployed)	--> Lender B
Fair/poor (any status)	--> Lender A

Optimal routing delivers+$875,928 in incremental value, a 33% RPA lift.



Current Revenue

$2.64M

Across 100k applications at $26.42 RPA

Optimized Revenue

$3.52M

At $35.18 RPA – Same traffic but this time, better routing

Incremental Gain

+875K

+33.2% RPA lift. Zero new traffic needed.

- Assumes approval rates are stable across routing changes. Real-time implementation would require live FICO + employment lookup and A/B testing in order to validate.

Smart routing is the easy part. Here's what to get right before going live.

A/B test before full rollout

Historical approval rates may not hold at new routing volumes. Run a holdout test routing 20% of traffic optimally before full deployment.

Lender capacity constraints

Routing everyone eligible to Lender C may exceed their volume appetite. Confirm capacity and volume commitments with each partner.

6,407 missing Employment Sector values

~6.4% null rate. Exclude from initial model or input as "unknown".

Approval rates will shift

Lender risk appetites change with macro conditions. Routing model needs periodic retraining. Recommendation: a monthly refresh.

Collect more signal over time

DTI is the best engineered variable here. We might consider adding: credit utilization, number of open accounts, income verification.

Fair lending compliance

Routing decisions must be audited for disparate impact. Ensure no protected class is systematically routed to worse outcomes.

Bottom Line/ In Summary

Route poor/fair FICO → A. Excellent/very good → C. Everything else is noise.

The 875k opportunity requires no new traffic. It just requires asking: are we sending each customer to the lender most likely to say yes at the highest bounty?

$35.18

Optimized RPA

+33%

Revenue Lift

$875K

Incremental Revenue